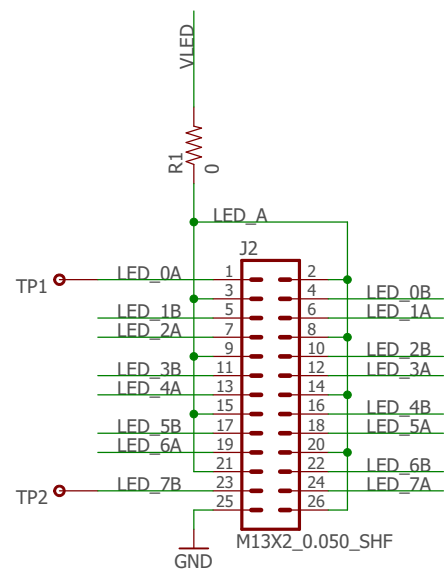
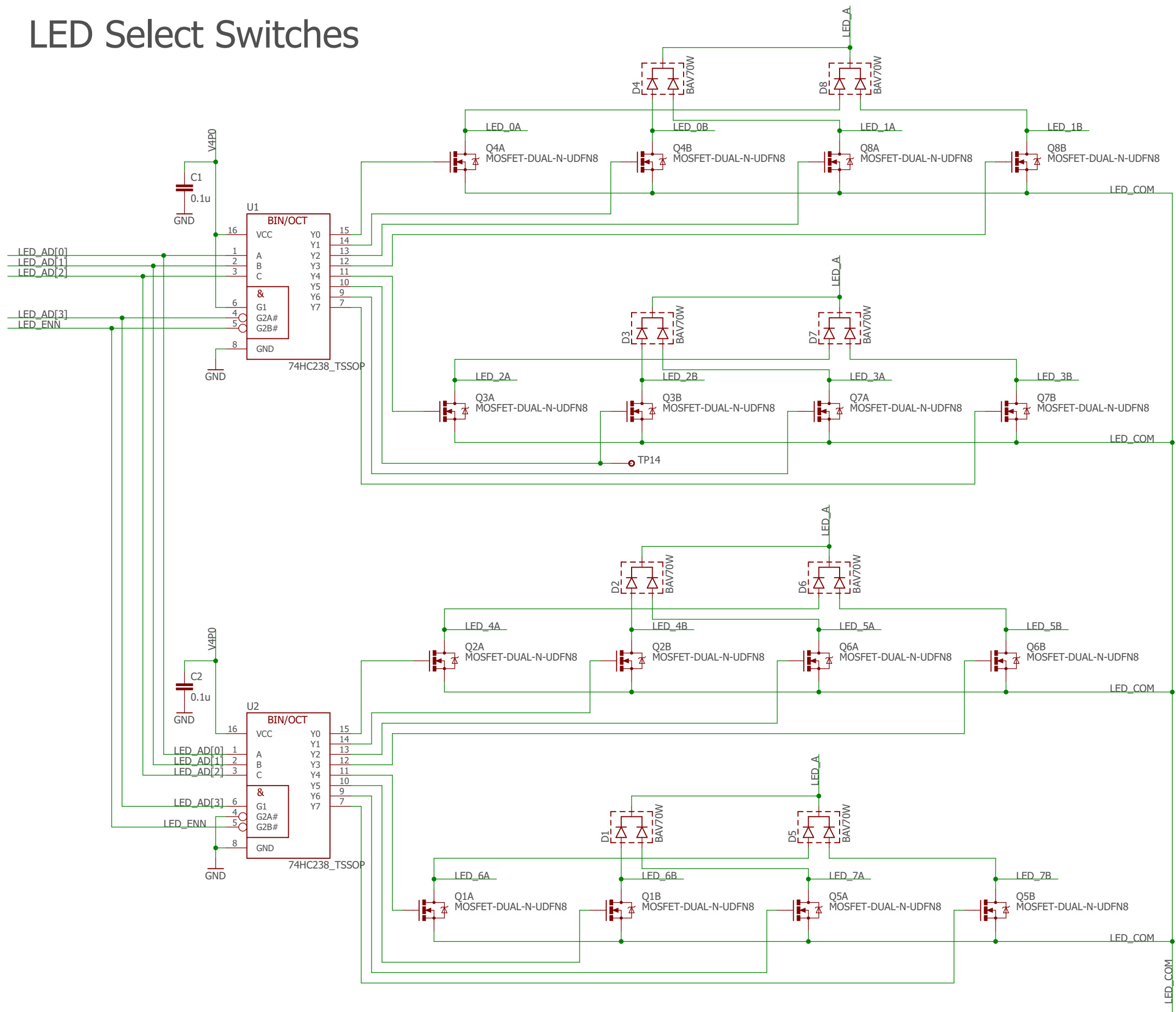
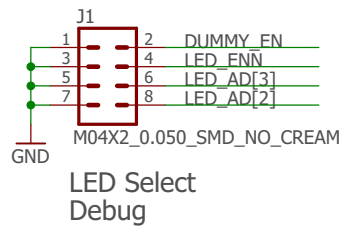


LED Select Switches



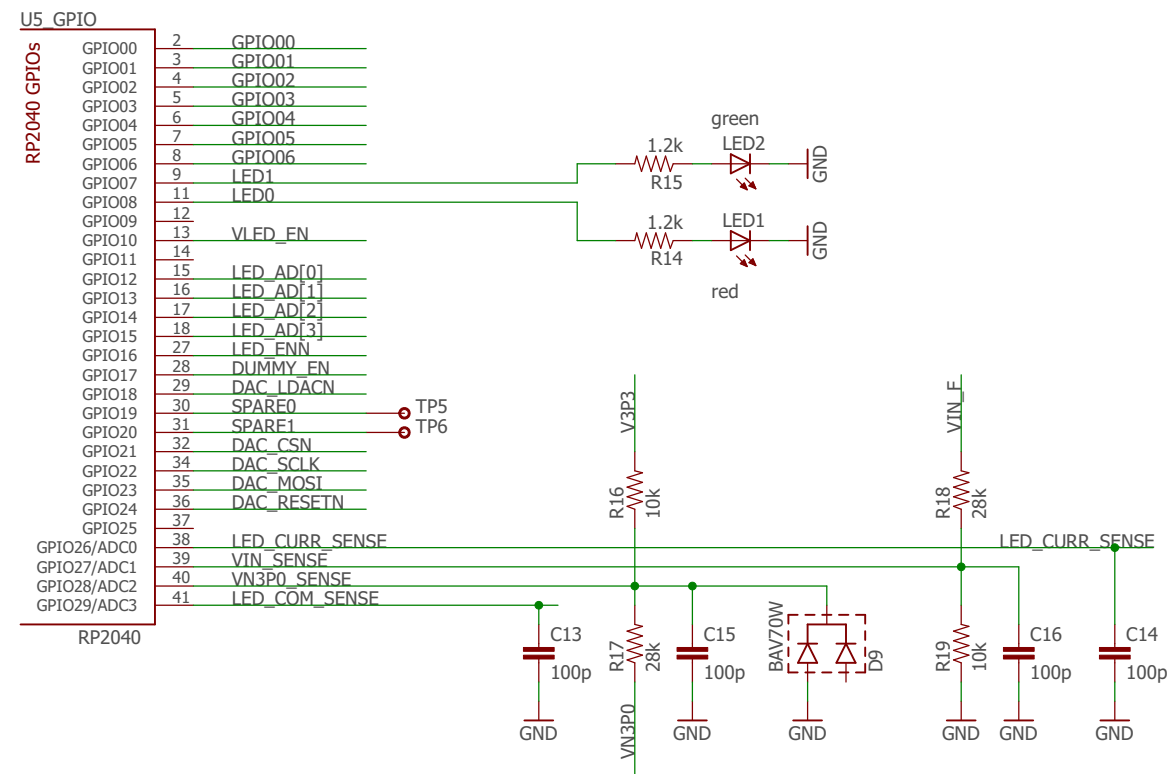
to Optode Bundle



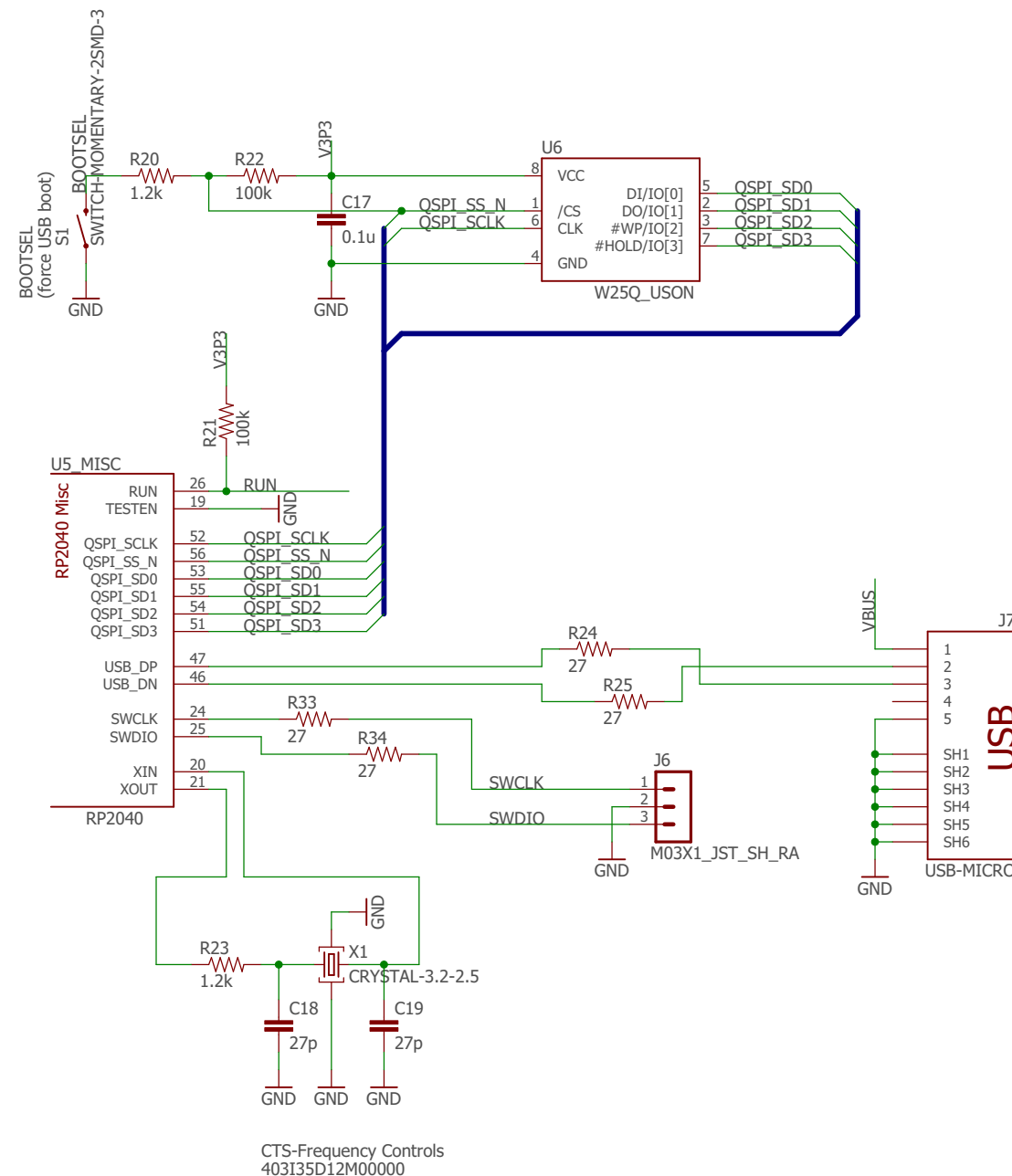
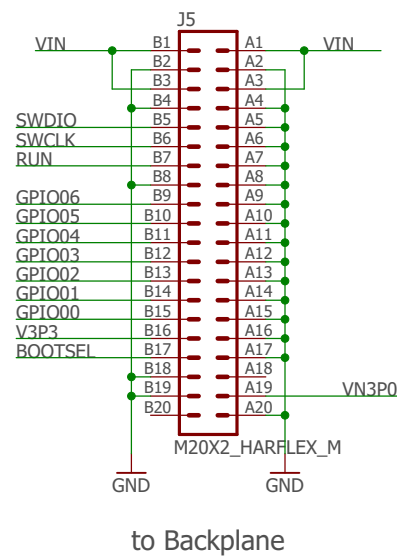
DAC / VCCS

The schematic diagram illustrates a DAC (Digital-to-Analog Converter) and VCCS (Voltage-Controlled Current Source) circuit. The core component is the AD5683R (U3), which is interfaced via a DAC SPI Debug header (J3). The DAC's output (VOUT) is connected to a resistor network (R2, R3) and a test point (TP3). The signal then passes through a buffer/amplifier stage (U4, OPA392) with feedback capacitors (C7, C8) and a resistor (R4). The output of the buffer is connected to a MOSFET driver (Q9B) and a current monitor (J4, U.FL). The MOSFET driver is connected to a MOSFET (Q9A) which drives an LED (VLED) through a resistor (R12). A dummy MOSFET (Q10A) is also shown for testing. Various capacitors (C3, C4, C5, C9, C10, C11, C12) and resistors (R5, R6, R7, R8, R9, R10, R11, R13) are used for timing and signal conditioning. The circuit is powered by V3P3 and V4P0, with ground connections throughout.

RP2040 / Backplane



Alt. Name (Firmware)	Alt. Name (FPGA)
RESERVED_PIN	SRCB_RESERVED
STATUS_SELECT_PIN	SRCB_SPARE
LED_EN_IN_PIN	SRCB_LED_EN
STEP_PIN	SRCB_STEP
BOARD_SELECTN_PIN	SRCB_SELECT
UART_RX_PIN	UART_FPGA_TO_SRCB
UART_TX_PIN	UART_SRCB_TO_FPGA



Voltage Regulators

